OneRhythm

Product Requirements Document

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A community-powered ECG mosaic for arrhythmia awareness

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Product Overview

Vision Statement

OneRhythm is a community-powered platform that unites arrhythmia patients through a collective heart mosaic. Users contribute de-identified ECG images that are visually composed into a growing heart silhouette, symbolizing solidarity. The platform returns personalized educational guidance based on the user's self-reported condition, generates questions to ask their cardiologist, and surfaces relevant medical advancements. It is not a diagnostic tool. It is a community, a visual statement, and an educational resource.

Mission

To reduce the isolation experienced by arrhythmia patients, aggregate awareness-raising data, and empower patients with accessible, multilingual educational resources—all built on a transparent, donation-funded, blockchain-verified infrastructure.

Target Users

- Primary: Patients diagnosed with ARVC, atrial fibrillation, ventricular tachycardia, and other arrhythmias.
- Secondary: Caregivers, family members, and patient advocates seeking education and community.
- Tertiary: Researchers and foundations interested in aggregated, de-identified arrhythmia data.

Regulatory Positioning

OneRhythm is designed as a general wellness and educational application. It does not interpret, analyze, or characterize ECG waveforms clinically. All educational output is driven by the user's self-reported diagnosis and symptom profile, never derived from the ECG image itself. The ECG is used solely as a visual contribution to the community mosaic. This positioning is designed to fall within FDA enforcement discretion for general wellness products per the 2026 updated guidance. Pre-launch regulatory counsel review is required before any public release.

Core Features

F1: ECG Image Contribution

Users upload an ECG image (photo, PDF, or exported file from a device). The image is processed through a PHI stripping pipeline that removes all embedded metadata, text overlays containing patient names, MRNs, dates of birth, and device identifiers. The cleaned image is

then processed through a waveform anonymization layer before being added to the mosaic database. The original upload is never stored.

Requirement	Detail	Priority
Supported formats	JPEG, PNG, PDF (single-page ECG exports)	P0
PHI stripping	OCR-based text detection and redaction of 18 HIPAA identifiers from image overlays; EXIF/metadata purge	P0
Waveform anonymization	GAN-based synthetic morphology perturbation (PrivECG-style) or Chronos-style synthetic generation to prevent biometric re-identification from waveform shape	P0
Consent gate	User must accept informed consent before upload; consent recorded on-chain	P0
Upload feedback	Visual confirmation showing the user's tile placement in the mosaic	P1

F2: Patient Profile & Self-Reported History

Users create a profile with their self-reported diagnosis, symptom history (physical and mental), treatment history, and personal narrative. This data drives all educational output. Validated instruments are used where applicable.

Requirement	Detail	Priority
Diagnosis selection	Structured picker: ARVC, AFib, VT, SVT, Long QT, Brugada, WPW, Other (free text)	P0
Symptom intake	Validated questionnaire (KCCQ-12 for heart failure symptoms, EQ-5D-5L for quality of life) plus free-text fields for personal narrative	P1
Treatment history	Structured fields: number of ablations, ICD implant (Y/N), medications, surgical history	P1
Mental health	PHQ-2 screening instrument plus free-text for emotional experience	P1
Data ownership	Users can view, edit, and delete all self-reported data at any time	P0

F3: Educational Guidance Engine

Based solely on the user's self-reported profile (never the ECG image), the system returns educational content. This is powered by an LLM with RAG retrieval against a curated medical literature corpus.

Requirement	Detail	Priority
Condition education	General information about the user's self-selected diagnosis drawn from peer-reviewed sources and major cardiology guidelines (AHA, ESC, HRS)	P0
Doctor questions	AI-generated list of questions the user might want to raise with their cardiologist, tailored to their reported symptoms and treatment history	P0
Medical advancements	Curated feed of recent research, clinical trials, and treatment developments relevant to the user's condition, updated monthly minimum	P1
Mental health resources	Links to support organizations, peer communities, and mental health guidance specific to living with arrhythmia	P1
Disclaimers	Every educational output must include a persistent, non-dismissible disclaimer: "This is educational information only. It is not medical advice. Consult your physician for clinical decisions."	P0
Multi-language support	LLM-based translation with medical terminology validation. Launch languages: English, Spanish, Italian, Greek, Portuguese, French, German	P1

F4: The Heart Mosaic

The emotional and visual centerpiece of OneRhythm. Every contributed ECG becomes a visual tile in a collectively growing heart silhouette. The mosaic is publicly viewable and serves as both a community artifact and an awareness tool.

Requirement	Detail	Priority
Tile generation	Each anonymized ECG waveform's visual characteristics (rhythm density, amplitude patterns) influence the tile's color, texture, and opacity within the heart outline	P0
Heart outline	A deterministic generative algorithm that fills a heart silhouette from center outward as contributions accumulate	P0
Real-time growth	The mosaic updates visibly as new contributions arrive; returning users see progress	P0

Interaction	Hover/tap on any tile to see: contribution date, self-reported condition type (no identifying info), and a count of total contributions from that condition category	P1
Embed/share	Shareable link and embeddable widget for advocacy use on external sites	P1
WebGL rendering	Three.js or equivalent for performant rendering at scale (target: 100,000+ tiles)	P0

F5: Blockchain Transparency Layer

A Solana-based ledger provides verifiable transparency for contributions, consent, and donations. Blockchain is infrastructure, not product surface—most users will never interact with it directly.

Requirement	Detail	Priority
Contribution ledger	Each de-identified ECG contribution receives a timestamped, immutable on-chain record (hash of anonymized data, not the data itself)	P1
Consent tracking	Smart contract records user consent actions with revocation capability	P1
Donation transparency	All donations tracked on-chain with public allocation visibility	P2
Off-chain storage	Actual ECG data stored encrypted off-chain (IPFS/Arweave or AWS S3 with client-side encryption); on-chain stores references only	P0
Right to deletion	Destroying off-chain data and encryption keys satisfies deletion requests; on-chain hash becomes meaningless	P0

Non-Functional Requirements

Privacy & Security

- All ECG processing (PHI stripping, waveform anonymization) occurs server-side in an isolated processing pipeline. No raw ECG data is persisted beyond the processing session.
- Encrypted at rest (AES-256) and in transit (TLS 1.3) for all stored data.
- User self-reported data is stored separately from anonymized ECG tiles with no technical link between the two after processing.

- Annual third-party security audit required post-launch.
- GDPR-compliant architecture from day one: right to access, right to deletion, data portability, explicit consent.

Performance

- ECG upload-to-tile processing: under 30 seconds end-to-end.
- Mosaic render: initial load under 3 seconds on modern hardware; progressive loading for tile interaction.
- Educational guidance generation: under 10 seconds from profile submission.
- 99.9% uptime target for the mosaic and educational engine.

Accessibility

- WCAG 2.1 AA compliance minimum.
- Screen reader support for all educational content and navigation.
- High contrast mode for the mosaic visualization.
- Keyboard navigation throughout.

Technical Architecture

System Overview

The system is composed of five primary subsystems: the user-facing web application, the ECG processing pipeline, the educational guidance engine, the mosaic rendering engine, and the blockchain transparency layer. Each operates as an independent service communicating via API boundaries.

Stack

Layer	Technology	Rationale
Frontend	Next.js 14+ (TypeScript), Tailwind CSS, Three.js	SSR for SEO, TypeScript for safety, Three.js for WebGL mosaic
ECG Processing	Python (FastAPI) on Railway/Modal	Python ecosystem for image processing, OCR (Tesseract), GAN models
Educational Engine	LLM API (Anthropic Claude) + RAG (vector DB)	High-quality generation with retrieval from curated medical corpus
Database	PostgreSQL (Supabase) + encrypted blob storage (S3)	Relational data for profiles, blob storage for anonymized tiles

Blockchain	Solana (Anchor framework)	Low transaction cost, high throughput, developer ecosystem familiarity
Off-chain Storage	IPFS/Arweave with client-side AES encryption	Decentralized, content-addressed storage for anonymized ECG data
Hosting	Vercel (frontend), Railway (backend), Supabase (DB)	Managed infrastructure appropriate for a lean team
Translation	LLM-based with medical terminology validation layer	Flexible, high-quality, supports rapid language expansion

Data Flow

User uploads ECG image → Server receives via TLS → PHI Stripping Pipeline (metadata purge + OCR text redaction) → Waveform Anonymization (GAN-based morphology perturbation) → Tile Generation (visual parameters extracted from anonymized waveform) → Tile stored in mosaic database → Hash recorded on Solana ledger → Original upload destroyed → User receives tile placement confirmation.

User completes self-reported profile → Profile stored in PostgreSQL (encrypted) → Educational Engine queries profile + RAG corpus → Returns condition education, doctor questions, advancement feed → Rendered in user's selected language.

Regulatory Compliance Requirements

FDA Positioning

Classification target: General wellness / educational tool, outside FDA SaMD jurisdiction.

Critical constraint: The application must never analyze, interpret, or characterize the ECG waveform for clinical or diagnostic purposes. All educational output must be derived from the user's self-reported profile, not from any computed analysis of the ECG image. The ECG contributes to the mosaic only.

Pre-launch requirement: Health tech regulatory attorney must review all user-facing copy, disclaimers, feature descriptions, and marketing materials before launch to confirm the application does not trigger SaMD classification.

HIPAA Compliance

- OneRhythm is not a covered entity and does not receive identified health information from healthcare providers. Users self-submit their own data voluntarily.

- Despite non-covered-entity status, the platform implements HIPAA Safe Harbor de-identification standards as best practice: all 18 identifiers stripped from ECG images, waveform-level anonymization to address biometric re-identification risk.
- Explicit informed consent obtained before any data is accepted, with all required elements: description of data collected, purpose, recipients, retention, and revocation rights.

GDPR Compliance

- Right to access: Users can export all their data at any time.
- Right to deletion: Off-chain data and encryption keys can be destroyed on request; on-chain hash becomes meaningless without the key.
- Data minimization: Only data necessary for the stated purposes (mosaic contribution, educational guidance) is collected.
- Consent: Granular, explicit, revocable consent for each data use (mosaic contribution, educational profiling, research aggregation).
- International transfers: Data processing within compliant jurisdictions; standard contractual clauses where required.

Disclaimer Requirements

The following disclaimer must appear persistently on all educational content screens, in the user's selected language, and must not be dismissible:

> *"OneRhythm is an educational resource and community platform. It is not a medical device. It does not diagnose, treat, cure, or prevent any disease. The information provided is for educational purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. Always consult your physician or qualified healthcare provider with any questions regarding a medical condition."*

Phased Development Roadmap

Phase 1: Foundation (Months 1–6)

- Engage health tech regulatory attorney for FDA classification guidance and disclaimer review.
- Engage HIPAA privacy consultant for de-identification architecture review.
- Build PHI stripping pipeline (metadata purge + OCR-based text redaction).
- Build waveform anonymization pipeline (GAN-based or Chronos-style synthetic generation).
- Design and implement patient intake forms with validated instruments (KCCQ-12, EQ-5D-5L, PHQ-2).

- Design the mosaic generative algorithm and heart silhouette rendering.
- Establish 501(c)(3) nonprofit structure.
- Apply for foundation grants (AHA, Robert Wood Johnson Foundation, rare disease foundations).
- Build core database schema and API boundaries.

Phase 2: MVP Launch (Months 6–12)

- Launch with: ECG contribution to mosaic, patient profile, educational guidance engine, doctor questions, medical advancement feed.
- English-only at launch; multi-language support in Phase 3.
- Deploy Solana contribution ledger for transparency.
- Build community through existing arrhythmia audience and partner organizations (SADS Foundation, ARVC registries).
- Intensive user feedback collection and iteration.
- Donation infrastructure live (Stripe for fiat, Solana for crypto).

Phase 3: Scale (Months 12–24)

- Multi-language rollout (Spanish, Italian, Greek, Portuguese, French, German).
- Research data partnership program for academic institutions (aggregated, de-identified data access).
- Embeddable mosaic widget for advocacy organizations.
- Media and awareness campaign tied to mosaic growth milestones.
- If regulatory clearance is obtained: explore AI-powered educational ECG context (future version, separate regulatory review).

Risk Matrix

Risk	Severity	Likelihood	Mitigation
FDA reclassification as SaMD	Critical	Low (with current design)	No ECG interpretation; regulatory attorney review pre-launch; all output profile-driven
ECG biometric re-identification	Critical	Medium	GAN-based waveform anonymization; privacy expert review; no raw ECG storage
Insufficient donation revenue	High	High	Diversified revenue: grants, research partnerships, corporate sponsors, merchandise

Liability from user acting on educational content	High	Medium	Persistent disclaimers; terms of service; professional liability insurance
Low initial adoption	Medium	Medium	Leverage existing arrhythmia audience; partner with SADS Foundation, ARVC registries
Blockchain complexity delays launch	Medium	Medium	MVP launches without blockchain; ledger added post-launch
International privacy law violations	High	Low	GDPR-compliant architecture from day one; off-chain data with deletion capability

Success Metrics

Metric	6-Month Target	12-Month Target	24-Month Target
ECG contributions to mosaic	500	5,000	25,000
Registered users	1,000	10,000	50,000
Monthly active users	300	3,000	15,000
Languages supported	1 (English)	3	7
Research data partnerships	0	2	5
Monthly recurring donations	$500	$5,000	$25,000
Mosaic embed installations	0	10	50

Explicitly Out of Scope (v1)

- AI-powered ECG interpretation or diagnostic analysis of any kind.
- Integration with electronic health record (EHR) systems.
- Telemedicine or provider-patient communication features.
- Wearable device real-time streaming integration (Apple Watch, Kardia, etc.).
- Clinical trial recruitment or matching.
- Premium/paid feature tiers (contradicts community mission for v1).

Appendix: Consent Language Template

The following is a template for the informed consent flow. Final language must be reviewed by regulatory counsel before use.

> *By contributing your ECG image to OneRhythm, you understand and agree that: (1) Your ECG image will be processed to remove all identifying information, including embedded text, metadata, and biometric waveform characteristics. (2) The anonymized visual representation of your ECG will become part of the OneRhythm Heart Mosaic, a publicly viewable community artwork. (3) Your self-reported health information will be used to provide you with educational content tailored to your condition. (4) Aggregated, de-identified data may be shared with research institutions for the purpose of advancing arrhythmia research. (5) You may withdraw your consent and request deletion of your data at any time. (6) OneRhythm is not a medical device and does not provide medical advice, diagnosis, or treatment.*

End of document.